Multiplexers & Demultiplexers

Multiplexer (Data Selectors)

- Multiplexer: Multiple input and single output.
- A multiplexer (MUX) is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line.
- A Multiplexer is also called a ***Data Selector***.
- The Multiplexer has several data input line and a single output line.

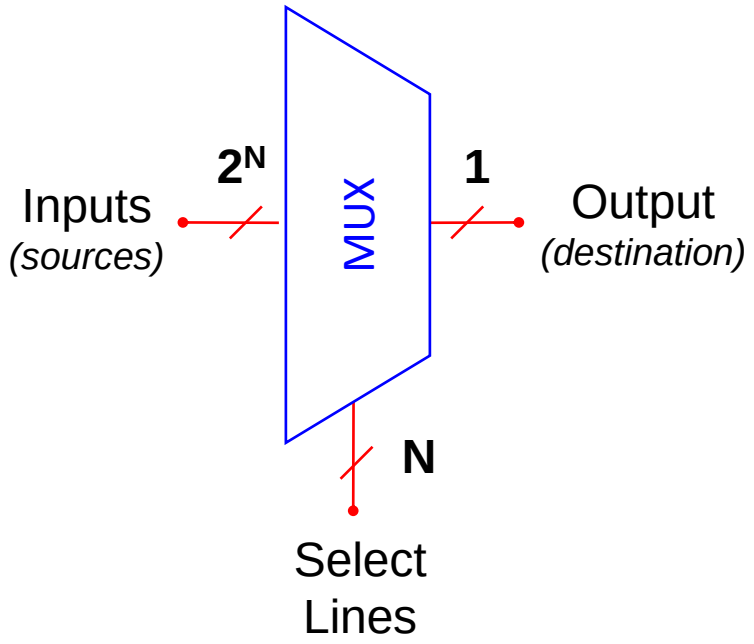
- MUX directs one of the inputs to its output line by using *selection line / control lines.*
- Multiplexer contains
  - *data inputs*
  - *selection inputs*
  - *a single output*
- Selection input determines the input that should be connected to the output.

- Normally, there are 2^N input lines and N selection lines whose bit combinations determine which input is selected.

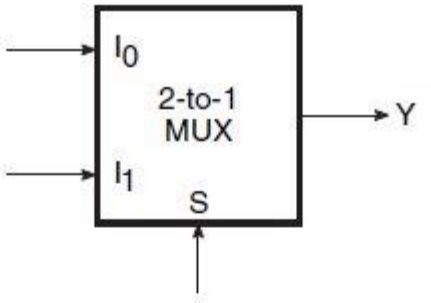
Type of Multiplexer

MUX	Select lines (N)	Input (2^N)	Output
2:1	1	2	1
4:1	2	4	1
8:1	3	8	1
16:1	4	16	1

Multiplexer Block Diagram



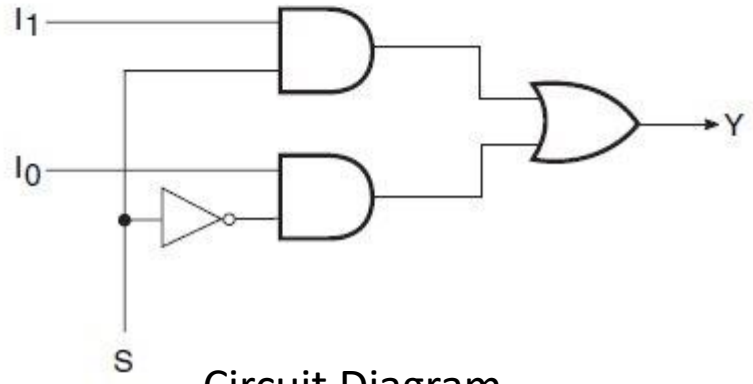
2:1 MUX



Block Diagram

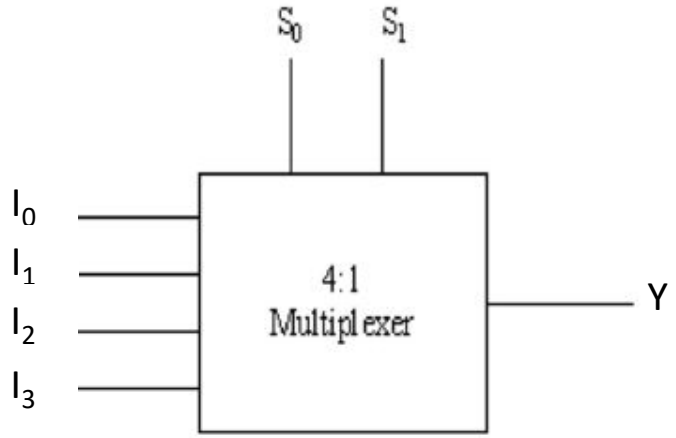
S	Y
0	I_0
1	I_1

Truth table

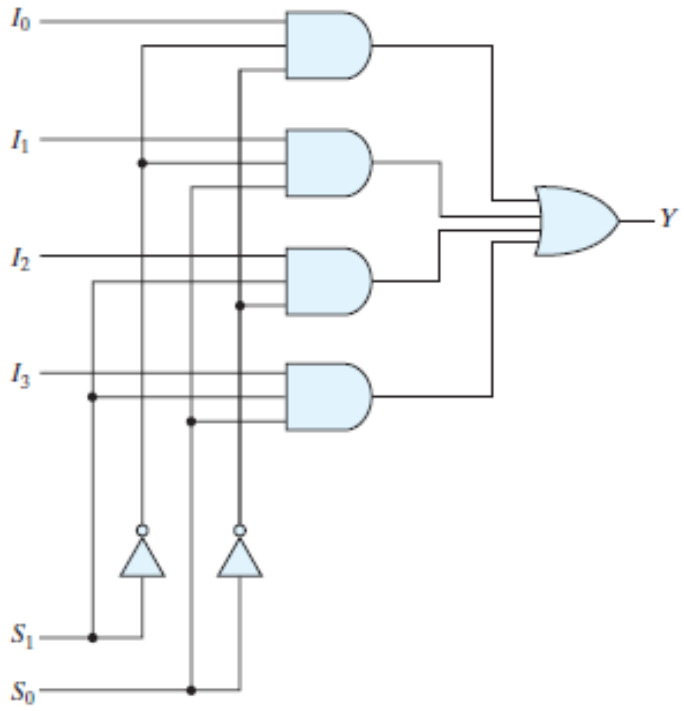


Circuit Diagram

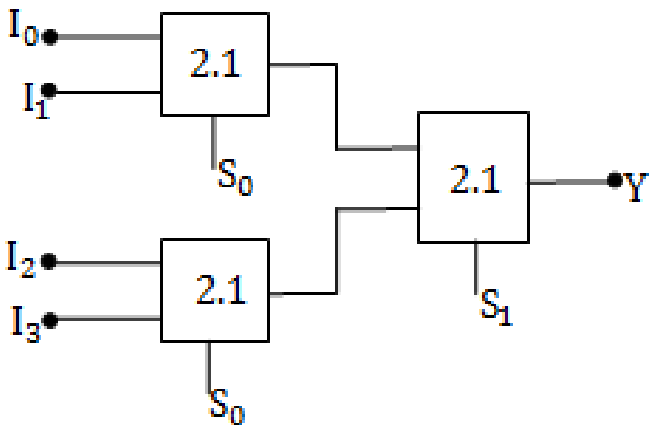
4:1 MUX



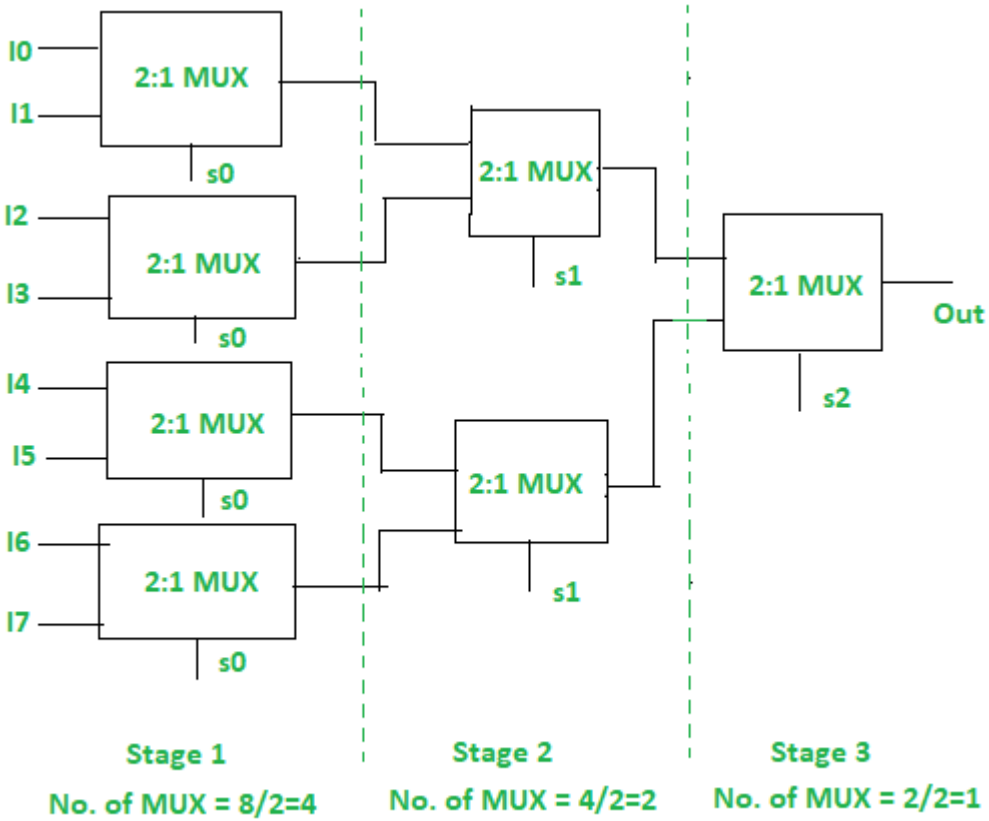
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



Implementation of 4:1 MUX using 2:1 MUX



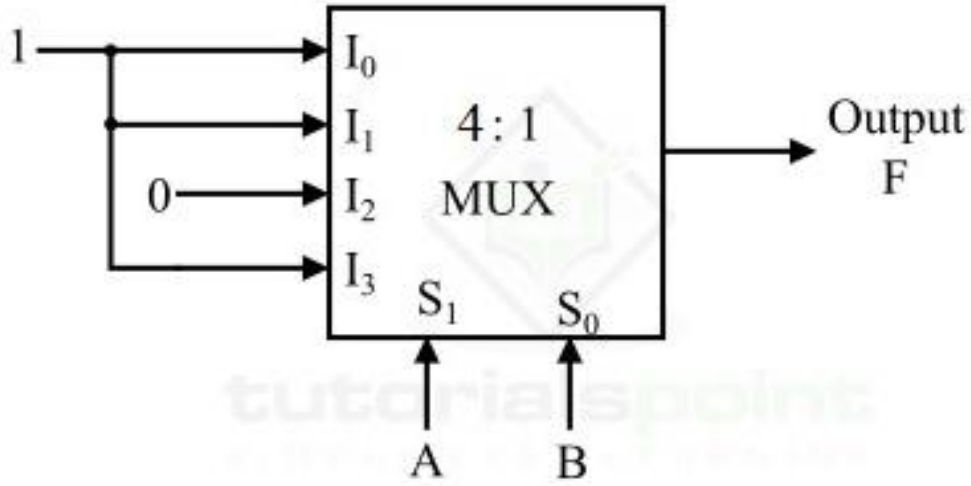
Implementation of 8:1 MUX using 2:1 MUX



Q1. Implement the given function, F(A,B)=∑m(0,1,3) using a 4:1 MUX

Ans:

Select Lines		Output	
$S_1 = A$	$S_0 = B$	Y	
0	0	1	I_0
0	1	1	I_1
1	0	0	I_2
1	1	1	I_3



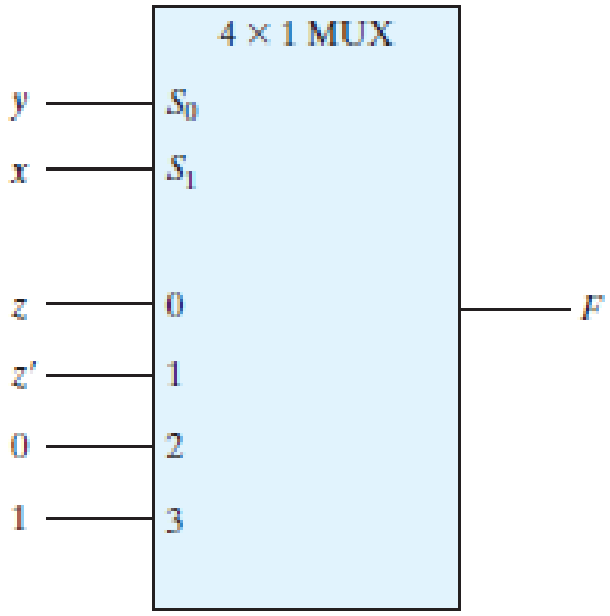
- Implement the following logical function using a 4:1 multiplexer,

  $F(x, y, z) = (1, 2, 6, 7)$

Ans:

x	y	z	F	
0	0	0	0	$F = z$
0	0	1	1	
0	1	0	1	$F = z'$
0	1	1	0	
1	0	0	0	$F = 0$
1	0	1	0	
1	1	0	1	$F = 1$
1	1	1	1	

(a) Truth table



(b) Multiplexer implementation

- **Exercise Problems:**

- Implement the following MUX
  a. 8:1 MUX using 2:1 MUX
  b. 16:1 MUX using 2:1 MUX

- Implement the following three variable logic functions using 4:1 multiplexer,

$$F(A,B,C)=\sum m(0,2,5,6)$$

De-Multiplexer

- The De-Multiplexer is a combinational logic circuit that performs the reverse operation of multiplexer (Several output lines, one input line).

- A DEMUX is a digital switch with a single input (source) and a multiple outputs (destinations).

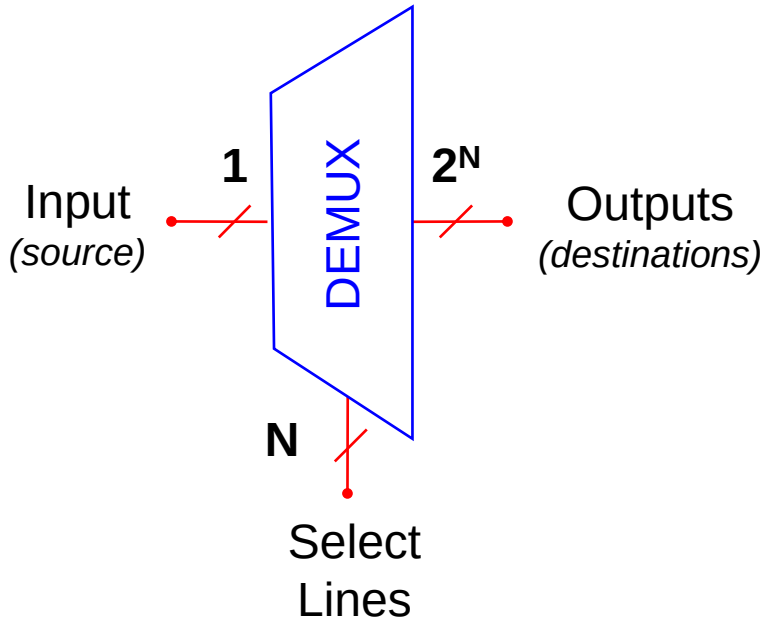
- A demultiplexer routes (or connects) the data input to selected output.

- The select lines determine which output the input is connected to.

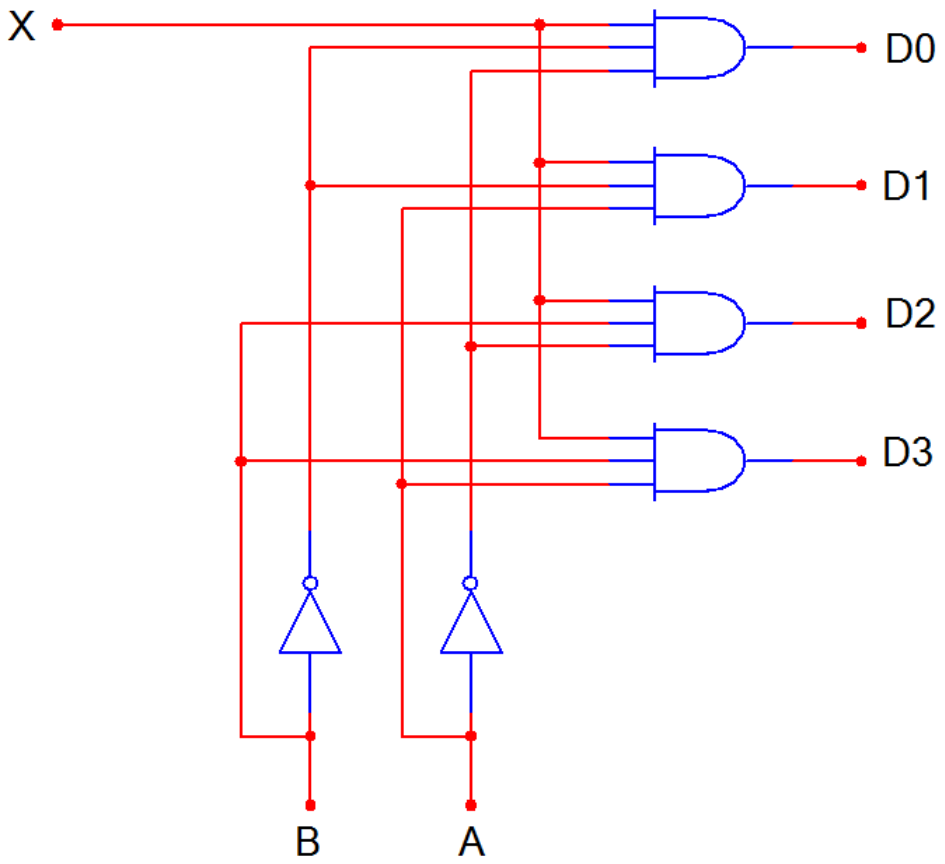
DEMUX Types

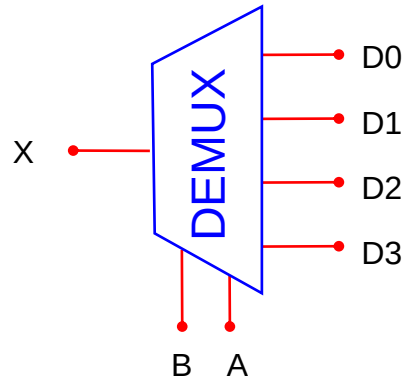
DEMUX	Select lines (N)	Input	Output (2^N)
1:2	1	1	2
1:4	2	1	4
1:8	3	1	8
1:16	4	1	16

Demultiplexer
Block Diagram



1:4 De-Multiplexer (DEMUX)





B	A	D0	D1	D2	D3
0	0	X	0	0	0
0	1	0	X	0	0
1	0	0	0	X	0
1	1	0	0	0	X